

SIMATS ENGINEERING
SAVEETHA SCHOOL OF ENGINEERING
 Department of Computer Science and Engineering
ASSESSMENT TOOL – 1
PSEUDOCODE WRITING

Course Code	DSA0108	Course Title	Object Oriented Programming with C++
CO Assessed	CO1 (BL3 & BL4)	Assessment	Tool – 1
Weightage	40%	Total Marks	100
Student Name	Sybil Liby A	Reg No	192421110
Section / Batch	Slot D	Date	
CO Statement	Apply object-oriented programming principles and C++ constructs such as data types, operators, and control structures to design and write structured OOP-based programs. (BL3 & BL4)		

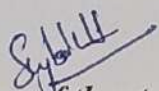
Total Questions: 10 Total Marks: 100

RUBRICS FOR EVALUATION

Criteria	Weightage	Level 4	Level 3	Level 2	Level 1
Problem Understanding	20%	All inputs correct	Minor missing	Partial	Incorrect
Algorithm Design	30%	Correct logic	Minor gaps	Partial	Wrong
Control Structures	20%	Efficient use	Minor issues	Basic use	Missing
Pseudocode Structure	10%	Well structured	Mostly clear	Hard to follow	Poor
Completeness	20%	All cases handled	Most cases	Basic	Incomplete

Code of Conduct:

I Sybil Liby A (192421110) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

ASSESSMENT - 1

Name : A. Sybil Litzzy

Reg no : 192421110

Course
code : DSA0108 - SLOTD

Course
Name : Object Oriented Programming with c++

1. A teacher wants to identify whether a student roll number is even or odd for seating arrangement. Complete the pseudocode by filling the missing parts to display the correct result.

START

Read Roll

If roll MOD 2 = 0 THEN

Print "Even"

Else

Print "Odd"

End If

STOP

B. A school wants to display whether a student has passed or failed based on marks. If the mark is 50 or above, display "Pass" otherwise display "Fail".

START

READ mark

If mark ≥ 50 Then

Print "Pass"

Else

Print "Fail"

End If

STOP

c. A library wants to calculate the fine for late return of books. If the number of late days is 0, no fine is charged; otherwise fine = late days $\times 5$. Complete the pseudocode by filling the blanks appropriately.

START

READ days

If days = 0 Then

fine = 0

Else

fine = days * 5

End If

Print fine

STOP

D. A weather station classifies temperature. If the temperature is above 35, print "Hot day"; If it is below 20, print "Cool day"; otherwise print "Pleasant Day".

START

READ Temp

If Temp > 35 Then

Print "Hot Day"

Else If Temp < 20 Then

Print "Cool Day"

Else

Print "Pleasant Day"

End If

STOP

E. A sports coach wants to compare the scores of two players & announce the winner. If both scores are equal print "Match Drawn".

START

READ score 1, score 2

```
If score 1 > score 2 Then
    Print "Player 1 Wins"
Else if score 2 > score 1 Then
    Print "Player 2 Wins"
Else
    Print "Match Drawn"
End If
STOP
```

2. A small grocery shop gives a discount to customers. If the purchase amount is ₹200 or more, the customer 10% discount; otherwise, no discount is applied. Construct the pseudocode to read the purchase amount & display the final bill amount.

```
START
READ amount
If amount >= 2000 Then
    discount = amount * 10/100
Else
    discount = 0
End if
bill = amount - discount
Print bill
STOP
```

A teacher wants to identify whether a student roll number is even or odd for seating arrangement. Write the pseudocode to read roll number & print odd or Even

```
START
Read roll
If roll Mod 2 = 0 Then
```


Print "Even"

Else Print "Odd"

End of

STOP

4. A tuition center needs to assign grades to students. If marks are 90 & above, assign Grade A; if 75 to 89, assign Grade B; if 50 to 74, assign Grade C; otherwise assign Grade D. Prepare pseudocode for grading process

START

READ marks

If marks ≥ 90 Then

Print "Grade A"

Else If marks ≥ 75 Then

Print "Grade B"

Else If marks ≥ 50 Then

Print "Grade C"

Else Print "Grade D"

End of

STOP

5. A bank wants to verify ATM withdrawal requests. If requested amount is less than or equal to available balance print "Transaction approved"; otherwise print "Insufficient Balance".

START

READ balance

READ amount

If amount $\leq$ balance Then

```
Print "Transaction Approved"  
Else  
Print "Insufficient Balance"  
End If  
STOP
```

6. A library wants to calculate the fine or late return of books. If the number of late days is 0, no fine is charged. Otherwise, $\text{fine} = \text{late days} \times ₹5$. Formulate the pseudocode to read the number of late days and display the fine amount.

```
START  
READ days  
If days = 0 Then  
    fine = 0  
Else  
    fine = days * 5  
End If  
Print fine  
STOP
```

A bus pass system checks passenger age. If the age is less than 12, print "Child Fare"; If the age is 60 or above, print "Senior citizen Fare"; otherwise print "Regular Fare". Design pseudocode for this system.

```
START  
READ age  
If age < 12 Then
```


Print "Child Fare"
Else If age ≥ 60 Then
Print "Senior Citizen Fare"
Else Print "Regular Fare"
End If
STOP

8. A sports coach wants to compare the scores of two players and announce the winner. If both scores are equal, print "Match Drawn". Write the pseudocode to read two scores and display the result.

START
Read Score 1
READ score 2
If score 1 $>$ score 2 Then
Print "Player 1 Wins"
Else If score 2 $>$ score 1 Then
Print "Player 2 Wins"
Else
Print "Match Drawn"
End If
STOP

9. A weather station records temperature. If the temperature is above 35°C , print "Hot Day". If it is below 20°C , print "Cool Day". Otherwise

print "Pleasant Day". Construct the pseudocode for this classification.

START

READ temp

If temp > 35 Then

Print "Hot Day"

Else If Temp < 20 Then

Print "Cool Day"

Else

Print "Pleasant Day"

End If

STOP

- 10 A number-checking system must identify whether an entered number is positive, negative or zero. Develop the pseudocode for this scenario.

START

READ num

If num > 0 Then

Print "Positive"

Else If num < 0 Then

Print "Negative"

Else

Print "Zero"

End If

STOP